

eco engineer

Introduction

Waste management has become an important challenge in educational institutions due to the increasing amount of waste generated by students every day. Most campus waste consists of food leftovers, paper, plastic wrappers, bottles, and other disposable materials. In many colleges, wet and dry waste are often disposed of in the same bin, making recycling and waste processing difficult. This leads to environmental pollution, poor hygiene, and increased landfill waste.

To address this issue, our team developed "Smarter Waste, Smarter Students", a smart waste management system that combines automated waste segregation with a reward-based approach. The system encourages students to dispose of waste responsibly while promoting environmental awareness and sustainable practices on campus.

Proposed System

The proposed system consists of a smart dustbin that automatically separates wet and dry waste into different compartments. The system uses an ESP32 microcontroller, sensors, and servo motors to identify and direct waste to the appropriate section. This reduces manual sorting efforts and improves recycling efficiency.

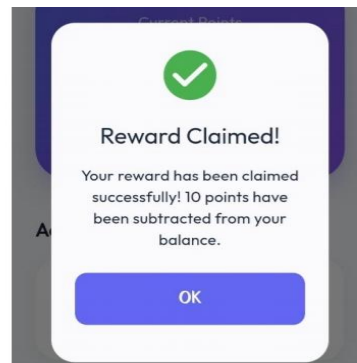
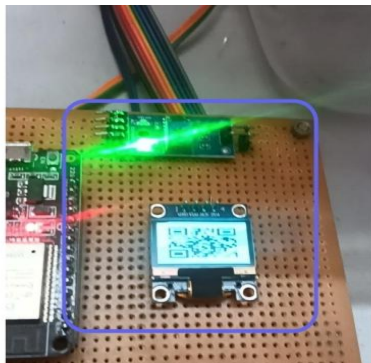
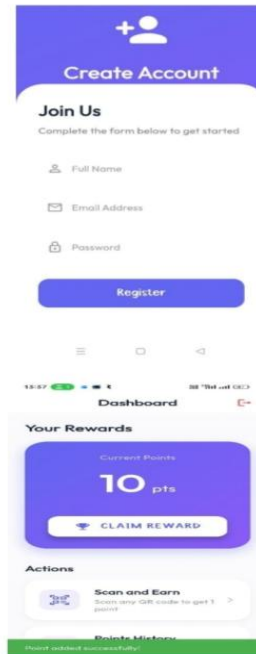
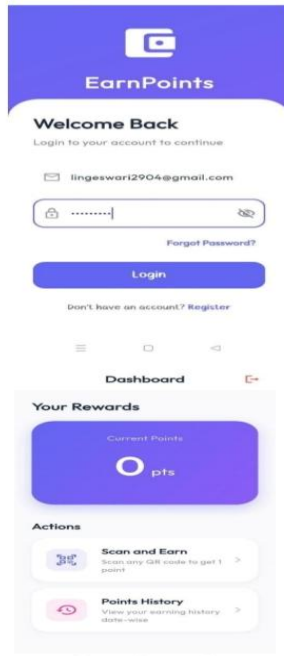
A QR code-based identification system is integrated to track student participation. When students dispose of waste correctly, their activity is recorded in a software platform connected through Firebase, a cloud-based database. As a reward for responsible behavior, students receive points and a chocolate, making the process engaging and motivating.

The software platform allows students to view their participation records and reward points, while administrators can monitor system usage and analyze waste management trends.

Key Features

- Automatic segregation of wet and dry waste.
- QR code-based user identification.
- Reward system with chocolate dispensing.
- Real-time data storage using Firebase.

- Student participation tracking.
- Administrative monitoring and reporting.
- Improved recycling and waste management efficiency.



Benefits and Impact

The system promotes proper waste segregation at the source, reducing contamination of recyclable materials. By introducing rewards, it encourages students to actively participate in maintaining a clean campus environment. The digital tracking system provides valuable data for monitoring waste disposal patterns and improving waste management strategies.

This project supports sustainable development by reducing environmental pollution, improving campus cleanliness, and creating awareness about responsible waste disposal practices. It can also be extended to schools, offices, shopping malls, and other public places.

Conclusion

Smarter Waste, Smarter Students is an innovative solution that combines smart technology and behavioral motivation to improve waste management practices. Through automatic waste segregation, digital monitoring, and a reward-based system, the project encourages responsible waste disposal and contributes to a cleaner, greener, and more sustainable environment.